

Conversion Log

Production Conversion Log

August 2025 · Girardot, Colombia

Submission reference	RTF0-310825
Reporting period	August 2025 (2025-08)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,775 kg/L
Density — PLUS-grade	0,859 kg/L
Production days observed	24 of 31 (7 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti

Managing Director

Date: _____

Executive Summary

Per-day kilogram-to-litre conversion log for **August 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	3 241 581,000 kg
Total EU-grade production	1 017 856,434 kg / 1 313 363,141 L
Total PLUS-grade production	1 274 935,420 kg / 1 484 208,869 L
Liquid-product yield (EU + PLUS) over input	70,73 %
Grand total volume (EU + PLUS)	2 797 572,010 L

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **459 429,842 kg**, steel scrap **255 554,307 kg**, water condensate **46 217,266 kg**, syngas **129 136,485 kg**, recorded losses **58 451,246 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — August 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,775 kg/L, PLUS 0,859 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-08-01	136 633,000	42 902,762	53 738,670	0,775	0,859	55 358,403	62 559,569	70,73 %	—
2025-08-02	119 304,000	37 461,456	46 923,059	0,775	0,859	48 337,363	54 625,214	70,73 %	—
2025-08-04	141 633,000	44 472,762	55 705,203	0,775	0,859	57 384,209	64 848,898	70,73 %	—
2025-08-05	130 972,000	41 125,208	51 512,161	0,775	0,859	53 064,785	59 967,591	70,73 %	—
Total	3 241 581,000	1 017 856,434	1 274 935,420	—	—	1 313 363,141	1 484 208,869	70,73 %	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-08-06	143 172,000	44 956,008	56 310,502	0,775	0,859	58 007,752	65 553,553	70,73 %	—
2025-08-08	131 567,000	41 312,038	51 746,179	0,775	0,859	53 305,855	60 240,022	70,73 %	—
2025-08-09	122 341,000	38 415,074	48 117,531	0,775	0,859	49 567,837	56 015,752	70,73 %	—
2025-08-11	138 200,000	43 394,800	54 354,981	0,775	0,859	55 993,290	63 277,044	70,73 %	—
2025-08-12	127 590,000	40 063,260	50 181,998	0,775	0,859	51 694,529	58 419,090	70,73 %	—
2025-08-13	143 533,000	45 069,362	56 452,486	0,775	0,859	58 154,015	65 718,843	70,73 %	—
2025-08-14	149 978,000	47 093,092	58 987,347	0,775	0,859	60 765,280	68 669,787	70,73 %	—
2025-08-15	136 103,000	42 736,342	53 530,218	0,775	0,859	55 143,667	62 316,901	70,73 %	—
2025-08-16	120 940,000	37 975,160	47 566,508	0,775	0,859	49 000,206	55 374,282	70,73 %	—
2025-08-19	142 636,000	44 787,704	56 099,690	0,775	0,859	57 790,586	65 308,137	70,73 %	—
2025-08-20	136 883,000	42 981,262	53 836,996	0,775	0,859	55 459,693	62 674,035	70,73 %	—
2025-08-21	140 075,000	43 983,550	55 092,431	0,775	0,859	56 752,968	64 135,542	70,73 %	—
2025-08-22	127 234,000	39 951,476	50 041,980	0,775	0,859	51 550,292	58 256,088	70,73 %	—
2025-08-23	138 250,000	43 410,500	54 374,647	0,775	0,859	56 013,548	63 299,938	70,73 %	—
2025-08-25	135 636,000	42 589,704	53 346,543	0,775	0,859	54 954,457	62 103,077	70,73 %	—
2025-08-26	147 297,000	46 251,258	57 932,892	0,775	0,859	59 679,043	67 442,249	70,73 %	—
2025-08-27	142 407,000	44 715,798	56 009,622	0,775	0,859	57 697,804	65 203,285	70,73 %	—
2025-08-28	137 665,000	43 226,810	54 144,563	0,775	0,859	55 776,529	63 032,087	70,73 %	—
2025-08-29	133 901,000	42 044,914	52 664,156	0,775	0,859	54 251,502	61 308,680	70,73 %	—
2025-08-30	117 631,000	36 936,134	46 265,057	0,775	0,859	47 659,528	53 859,205	70,73 %	—
Total	3 241 581,000	1 017 856,434	1 274 935,420	—	—	1 313 363,141	1 484 208,869	70,73 %	

Non-production days (7): 2025-08-03, 2025-08-07, 2025-08-10, 2025-08-17, 2025-08-18, 2025-08-24, 2025-08-31. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during August 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform `product_densities` table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to August 2025 are:

- **EU-grade (DEV-P100-EU)** — $\rho_{\text{EU}} = 0,775 \text{ kg/L}$
- **PLUS-grade (DEV-P100-PLUS)** — $\rho_{\text{PLUS}} = 0,859 \text{ kg/L}$

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform `product_densities` table — initially loaded by Alembic migration `0005_seed_product_densities.py` and extended with per-month effective rows by `0014_monthly_densities.py`. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu    NUMERIC GENERATED ALWAYS AS (eu_prod_kg / 0.775) STORED,
litres_plus  NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.859) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix 000000000000). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

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